

TactiHelm: Tactile Feedback in a Cycling Helmet for Collision Avoidance

Dong-Bach Vo
dong-bach.vo@glasgow.ac.uk
School of Computing Science,
University of Glasgow
Glasgow, United Kingdom

Julia Saari
2207399S@student.gla.ac.uk
School of Computing Science,
University of Glasgow
Glasgow, United Kingdom

Stephen Brewster
stephen.brewster@glasgow.ac.uk
School of Computing Science,
University of Glasgow
Glasgow, United Kingdom

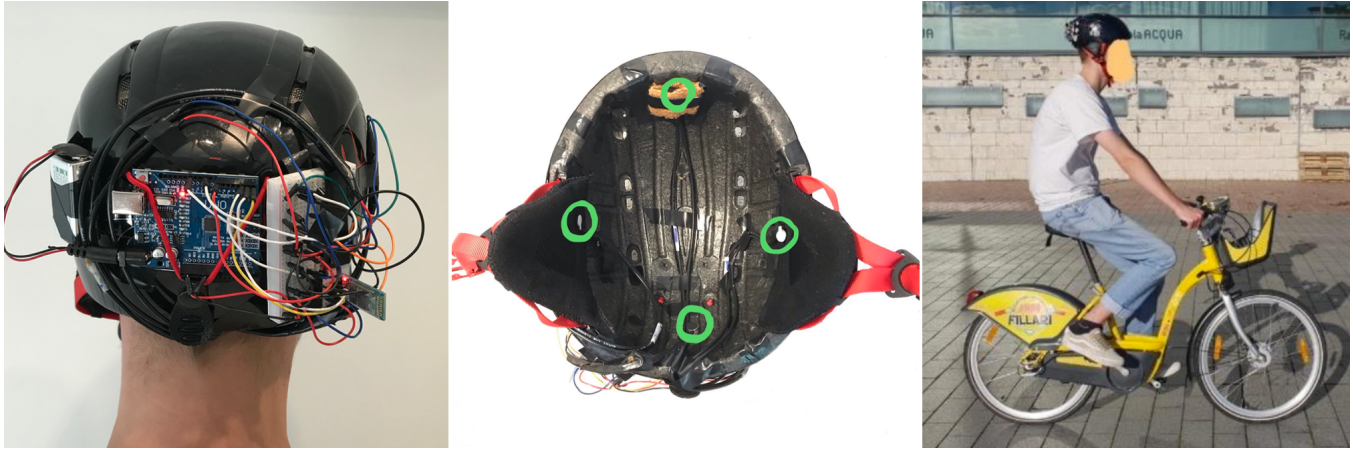


Figure 1: The TactiHelm prototype. It has 4 tactile actuators located at each of cardinal directions inside the helmet. The actuator is driven by an Arduino microcontroller that can be controlled remotely via Bluetooth. The rightmost picture shows a participant wearing TactiHelm during the evaluation.

ABSTRACT

This paper introduces TactiHelm, a helmet that can inform cyclists about potential collisions. To inform the design of TactiHelm, a survey on cycling safety was conducted. The results highlighted the need for a support system to inform on location and proximity of surrounding vehicles. A set of tactile cues for TactiHelm conveying proximity and directions of the collisions were designed and evaluated. The results show that participants could correctly identify proximity up to 91% and directions up to 85% when tactile cues were delivered on the head, making TactiHelm a suitable device for notifications when cycling.

KEYWORDS

haptic feedback, cycling, haptic helmet, cycling helmet, haptic notifications

ACM Reference Format:

Dong-Bach Vo, Julia Saari, and Stephen Brewster. 2021. TactiHelm: Tactile Feedback in a Cycling Helmet for Collision Avoidance. In *CHI Conference on Human Factors in Computing Systems Extended Abstracts (CHI '21 Extended Abstracts)*, May 8–13, 2021, Yokohama, Japan. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3411763.3451580>

1 INTRODUCTION

With 2160 fatal injuries from traffic collisions in 2018, cyclists accounted for 8% of all road deaths in the EU [1]. While enhancing the cycling experience through technology to support secondary tasks such as navigation or social cycling has been a focus of most research in the HCI community [5, 10–14, 16], little work has looked at improving cyclist safety from other road users.

The attentional demand of cycling in urban areas is high. Cyclists need to focus their visual and auditory senses to navigate and constantly evaluate risks in their surroundings to avoid potential collisions [17, 18]. Tactile cues are a potential alternative as they do not disrupt visual or auditory attention. Previous research has shown that they can convey information in a non-intrusive way, for example, giving navigation information through vibration on the handlebars of the bike [11]. Other research has shown that tactile feedback can even reduce distraction while cycling [18]. However, little research has investigated how tactile feedback can inform cyclists about surrounding obstacles.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than ACM must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

CHI '21 Extended Abstracts, May 8–13, 2021, Yokohama, Japan

© 2021 Association for Computing Machinery.

ACM ISBN 978-1-4503-8095-9/21/05...\$15.00

<https://doi.org/10.1145/3411763.3451580>

In this paper, we introduce TactiHelm, a cycling helmet which uses tactile feedback to convey information to a cyclist's head. It indicates the location of potential dangers such as other vehicles nearby that could cause collisions. In a final system, it could be used with a bicycle radar which can detect the locations and distances of vehicles in the environment. These devices are small enough to be used on bikes, e.g. the Garmin Varia [4]. However, they mostly give visual and audio feedback, e.g. from a bike computer, which could potentially be distracting. TactiHelm could provide the information to the rider's head.

Two studies were carried out to begin the design process for TactiHelm. An online survey was conducted to understand cyclists' needs, concerns and behaviour related to safety on the road. We then designed and evaluated a set of tactile cues in a cycling helmet to inform riders of the distance and direction of vehicles in their surroundings. Results show that identifying distance was easier than identifying direction and that simpler tactile cues were better at indicating obstacles at a further distance.

2 RELATED WORK

Previous research has investigated tactile feedback for cycling. Most of the efforts have focused on tactile aids to inform cyclists about about direction and distance for navigation tasks [5, 10–14]. For instance, HaptiMoto uses a haptic vest to indicate upcoming turns direction on the users' back [13], Tactile wayfinder conveys directional cues through a belt worn by the cyclist [5]. However, these require users to wear non-cycling attire to benefit from tactile cues. Tacticycle gives cyclists compass information through vibration in the bike handlebars to help them reach their destination [11].

Other research has proposed tactile aids for collision warnings. Combining feedback locations such as handlebars, saddle and helmet, and modalities such as audio, visual and tactile, Matvienko *et al.* have examined feedback techniques to increase children's awareness of potential collisions when cycling [9]. The researchers found that unimodal feedback was the most efficient to alert children of immediate danger, tactile feedback was especially suited for conveying spatial cues to child cyclists. However, little research has explored tactile support for collision avoidance in cycling. Our research aims at exploring novel ways to inform cyclists of potential collisions using tactile notifications.

Adding technology to cycling helmets has received interest in the human-computer interaction community. Jones *et al.* have envisioned a smart helmet capable of detecting cyclists' head input to communicate their intentions through lights to other road users [6]. LumaHelm allows its wearer to transform data into a light representation to share with surrounding persons [15]. In the same vein, Open Heart Helmet displays heart rate data to the following cyclist to improve social cycling experiences [16]. Audio feedback in cycling helmets has been studied to warn children of dangers [9]. Although researchers have explored visual and auditory media to communicate information in cycling helmets, no research has yet investigated haptics.

We are interested in exploring the ability of cycling helmets to deliver tactile messages to cyclists. They are commonly worn in many countries, and are legally required in some, so provide a

good platform for delivering messages. To address the overlooked problem of collision avoidance, we have designed and evaluated a set of tactile cues for the head that can inform cyclists about the direction and distance of potential obstacles. Our investigation starts with an exploratory survey about cyclists' safety concerns to inform the design of such a helmet.

3 EXPLORATORY SURVEY

To identify meaningful pairings between information and tactile cues, and to inform the design of a tactile helmet prototype, an online survey on cycling safety was conducted. The survey was composed of 15 questions which explored the safety equipment participants used, strategies for awareness of other road users and the challenges of cycling safety. Finally, the survey invited participants to comment on equipping a cycling helmet with tactile feedback. The main focus was to identify the most important information cyclists wanted to know about other road users.

The survey was answered by 51 respondents aged from 22 to 68 years old ($M = 33.56$, $SD = 13.55$). The participants were located in the UK (56.86%), in the US (23.53%), and in Finland (13.73%). Out of 51 participants, 64.7% reported spending 3 or more hours cycling per week, 15% spent less than one hour per week cycling. While 32.54% of the participants cycled mainly in urban environments, 27.78% cycled in the countryside, 27.78% in parks, 9.52% in the mountains, and 6.35% in other environments. 72% of respondents regularly shared the road with motorised vehicles.

Participants reported several strategies for staying safe on the road. Visual awareness was the most common. Strategies such as looking out for cars by performing shoulder checks and scanning the surroundings was mentioned by 35 participants. For example, one participant summarised his strategy with: "[I stay alert of traffic on the road by] turning my head frequently, looking well ahead to predict flow of traffic and dangers". Mentioned by 27 participants, auditory awareness, such as listening for the sound of approaching vehicles or car engines was the second most prominent safety strategy. For example, one participant said "Look around me, listen carefully if there's someone behind me". Other strategies included making hand signals, defensive cycling, use of bike mirrors or cameras.

Participants were also questioned about their use of helmets and other safety equipment or technologies. More than 82% reported wearing a helmet while cycling on a regular basis. In addition to a helmet, respondents reported using cycle lights (26.01%) and reflective gear (13.87%). Several participants also reported using other technologies including bicycle computers (9.25%), GPS navigators (9.83%) and heart rate monitors (9.25%).

Sharing the road with other vehicles was the biggest safety concern among participants (80%). They saw other vehicles as a significant challenge to cycling safety. They were concerned about distracted or aggressive drivers, or cars encroaching on cyclists space: "Cycling in the same lane as cars feels unsafe", "The inconsiderate behaviour of motorists impatient to get past [is a challenge to safety]". Unexpected passing or encroaching cars were seen as hazardous: "Times when I don't see a car, can be alarming to have car pull by also times when vehicles don't see me and/or pass very

closely”. Finally, participants were asked to rank what they thought was the most important information for a tactile device to communicate about other vehicles on the road. The most important aspects were proximity of approaching vehicles (ranked most important by 35.3%) and location of the approaching vehicle (ranked most important by 13.7%).

As collisions with cars account for the majority of cycling road accidents [1], a key strategy to increase road safety in cyclists could be to help them be more aware of other vehicles on the road. To address this, we propose TactiHelm, an interactive tactile cycling helmet to increase cycling safety when sharing the road with other vehicles. The aim of this device is to alert the cyclist of approaching vehicles on the road using vibration cues in the helmet. As cyclists rely on vision and audition during cycling, vibration cues were selected as they can be used to deliver information without overloading the other senses. Furthermore, the cycling helmet is an ideal location for tactile cues as it is widely worn and always in contact with the head.

4 TACTIHELM, A TACTILE CYCLING HELMET FOR PROVIDING AWARENESS OF APPROACHING VEHICLES

Following the survey outcomes, we investigated tactile cues to convey proximity and direction of approaching vehicles. TactiHelm is a cycling helmet equipped with four 3 volt ERM coin cell tactile actuators with a diameter of 10mm attached to the inside lining at the front, back, left and right side of the helmet (Figure 1). Similar to previous approaches where the location of a tactile stimulus provides directional information for navigation [7], TactiHelm indicates one of the four cardinal directions by setting off one of the actuators around the cyclist’s head. The actuation results in a tactile cue perceivable on either the forehead, the left temple, the right temple or the back of the head.

The vibrotactile actuators in TactiHelm are driven by PWM signals from 0 to 3V generated by an Arduino Uno micro-controller [8]. TactiHelm is equipped with a Bluetooth module which allows remote communications. A custom Android application was developed to allow the researchers to trigger a stimulus on demand. The application includes a button to trigger each of the 8 cues and was run on an Android mobile phone.

To deliver rapid and discernible levels of information about proximity, we designed two different tactile patterns by varying the occurrence and the duration of vibration as previously proven performant for coding distance [3, 14]. While a single vibration of 500ms alerted of a vehicle located at a far distance, two vibrations of 200ms separated by 200ms were used to warn cyclists of a vehicle at close distance. The cues were also inspired by those provided by the Garmin Varia radar, which first alerts a rider to a vehicle approximately 140m behind them (the Varia uses a sound and a visual display from the bike computer), and then again when the vehicle is approximately 10m behind them. A future version of TactiHelm would get the data from the radar system, but for our studies we simulated this via Bluetooth.

5 USER STUDY: TACTILE CUES PERCEPTION AND IDENTIFICATION

We conducted an initial user study to investigate whether these tactile cues delivered to the head can be accurately perceived and identified while cycling.

5.1 Experimental procedure

The study was conducted in Helsinki on cycling tracks and open gravel roads in wide areas with no obstacles. The locations had no traffic or pedestrians. Each track ridden during the study was approximately 20 meters long. To standardise the equipment and ensure safety, 7 participants cycled on a Helsinki city bike which can only be ridden up to 15 kmh. These bikes also allow participants to adjust for height for cycling comfortably and safely. Three participants used their own bikes. The experimental procedure began with the participant putting the TactiHelm on.

Vibration cues were triggered by the researcher using a mobile phone connected to the helmet via Bluetooth. For each stimulus, participants were asked to state the direction of the cue and the proximity level associated to the cue (to ensure they could map the cue to its meaning). Before the experiment started, participants were presented with the 8 stimuli as part of a training phase. During this phase, they could ask questions to ensure they understood the cues and what they had to do for the study. For each trial in the experimental phase, a tactile cue was presented to the participants while they were cycling the track. Upon reaching the end of the track, participants were asked to verbally report the direction and the proximity level. The experimenter logged the identification success of each property for each trial data. In total, each participant performed two series of 8 trials. Trials were randomised for each series. At the end of the experiment, participants were interviewed about their experiences with TactiHelm.

To summarise, two independent variables, *Location* of the actuators and the tactile *Pattern* were studied. The *Location* could either be *Front*, *Left*, *Right* or *Back* and the *Pattern* either *Single* or *Double*. Two dependent variables were measured, the identification success of the *Direction* and the identification success of the *Proximity*.

Ten volunteers aged between 18 and 57 years old ($M = 29$, $SD = 11.66$) participated to the study. The study was approved by our institutional ethics committee.

5.2 Results

The overall mean accuracy for identifying *Proximity* was 91% ($SD = 0.287$). The mean accuracy for identifying *Proximity* from a *Single* vibration was 93.8% ($SD=0.243$) and 88.1% ($SD=0.325$) from a *Double* vibration. It was 82.5% ($SD=0.382$) for identifying *Proximity* from the actuator located on the *Back*, 95% ($SD=0.219$) from the one on the *Front*, 90% ($SD=0.302$) from the one on the *Left* and 97.5% ($SD=0.191$) from the one on the *Right* side of the helmet.

The overall mean accuracy for identifying *Direction* was 85% ($SD = 0.358$). The mean accuracy reached 87.5% ($SD=0.381$) for identifying *Direction* from a *Single* vibration and 82.5% ($SD=0.332$) from a *Double* vibration. It reached 67.5% ($SD=0.471$), 86.2% ($SD=0.347$), 88.8% ($SD=0.318$) and 97.5% ($SD=0.157$) for a stimulus coming from the actuator located on the *Back*, *Front*, *Left* and *Right* side of the

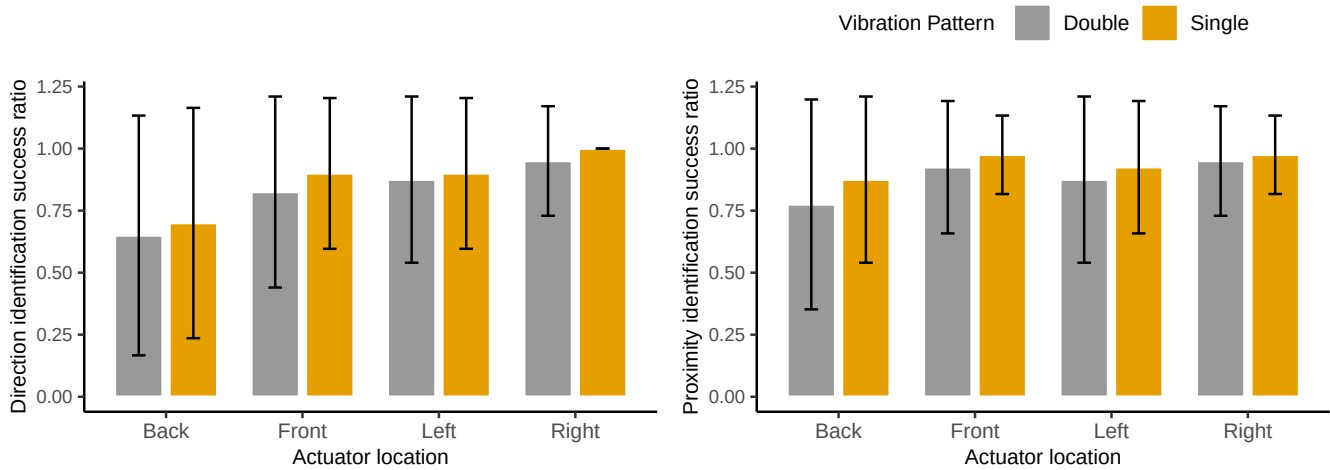


Figure 2: Identification accuracy ratio by actuator location and vibration pattern. The error bars denote SD values.

helmet. The results for each location of actuators on the helmet and each tactile pattern are shown in Figure 2.

A logistic regression analysis was performed using *Location* and *Pattern* as predictors for the identification of the *Direction* and *Proximity*. The odds ratios of each outcome are presented in Table 1. A Wald chi-squared test on the model coefficients revealed an overall significant effect of *Location* on *Distance* ($\chi^2(3) = 1.7, p < .01$) and *Proximity* ($\chi^2(3) = 10.1, p < .05$). This indicates that the actuator location had a significant effect on the identification of the directions and the proximity levels. No other significant effect was found.

When participants were asked about their experiences of cycling with TactiHelm, 7 of them felt comfortable. Half of the participants, who reported discomfort while wearing TactiHelm, mentioned difficulties in identifying directional cues. Most participants found the tactile cues from the helmet to be a good method of receiving information. Only a single participant reported being distracted and having to put more effort into identifying the cues. Finally, 7 out of 10 participants did not have any difficulty in remembering the meaning associated to the cues. They reported that short bursts felt more “imminent” or “urgent” than longer vibrations. It is worth mentioning that 1 participant expressed a preference for the inverse mapping. Finally, 2 participants wanted the distance information to be more accurate by using either more levels or a continuous mapping.

5.3 Discussion

The results show that tactile feedback to the head can deliver direction and distance cues very accurately, indicating that TactiHelm was a successful concept and would be useful to cyclists. The results indicate that actuator location played a significant role in the identification of direction and proximity level. Information in the tactile cues delivered by the actuator at the back of the helmet were not identified as well as those delivered by the other actuators, which differs from previous research results on tactile perception on the head [2]. This may be due to a poorer fit of the helmet in

that area, for example due to hair. In addition, the information from the cues delivered to the right side of the helmet was better perceived than from the cues delivered to the front and left side of the helmet, as confirmed by the calculated odd ratios. Although occipital region is suitable for tactile warnings, previous research has found the frontal region of the head to be the most suitable location for delivering tactile cues [2]. This may be explained by the fact that our helmet did not fit all participants perfectly. Further investigation is needed to confirm our initial findings. There are hints in our data (Figure 2) that single longer vibrations may be easier to perceive than short repetitive vibrations on the head while cycling. Additional research is needed to probe this further.

Participants found tactile cues easy to distinguish and meaningful, and TactiHelm was seen as useful. Furthermore, they also thought that short repetitive vibrations should be associated with urgency. The identification of information were accurate, making tactile stimulation on the head a viable support for communicating information to cyclists. Participants seemed to be comfortable with using TactiHelm and were not distracted by the tactile cues. Nevertheless, half of the participants suggested further improvements, such as increasing the distance precision. The feedback from participants suggests that the mapping between the current tactile pattern and proximity could be improved. Finally, only 4 actuators were used in this study. Increasing the number of actuators could allow to convey more complex directional cues to cyclists. We will investigate this further.

6 CONCLUSION

This paper has introduced TactiHelm, a helmet that can inform cyclists about potential obstacles around them using tactile feedback. Results showed that participants could understand distance and direction cues well, and liked the concept. To inform the design of TactiHelm, a survey was conducted. The outcomes highlighted “vehicles on the road” as the biggest concern for cyclists’ safety and the desire for a support system to inform on location and proximity

Table 1: Results of the logistic regression analyses: odds ratio for the identification of the directions and the proximity levels with the location of actuators and the tactile patterns.

	Identification of Direction		Identification of Proximity	
	Odds Ratio (95% CI)	p	Odds Ratio (95% CI)	p
Intercept	2.60 (1.74-4.79)	0.00139	7.09 (3.45-16.27)	0.0000006
Front	3.04 (1.41-6.97)	0.00590	4.09 (1.38-15.06)	0.01769
Left	3.84 (1.41-6.97)	0.00170	1.93 (0.77-5.13)	0.17008
Right	19.04 (5.37-121.50)	0.0000969	5.54 (1.71-24.88)	0.00956
Near	0.65 (0.33-1.23)	0.19015	0.48 (0.21-1.07)	0.07893

of surrounding vehicles. A set of tactile cues for TactiHelm conveying proximity and direction were designed and evaluated. We found that tactile notifications on the head were a viable support for communication when cycling, and the location of the actuators had a significant effect on identification of the tactile cues they presented. We have already begun to design new tactile cues to carry richer information than the ones presented here. With TactiHelm, our ongoing research has the potential for improving safety and communication to cyclists, and hopefully bringing more people on bikes.

ACKNOWLEDGMENTS

This research is funded by the European Union's Horizon 2020 research and innovation programme (PRESTIGE, grant agreement #761112).

REFERENCES

- [1] Dovič Admaité-Fodor and Graziella Jost. 2020. *How safe is walking and cycling in Europe?* Technical Report 38. European Transport Safety Council, Brussels, Belgium. <https://etsc.eu/how-safe-is-walking-and-cycling-in-europe-pin-flash-38/>
- [2] Victor Adriel de Jesus Oliveira, Luciana Nedel, Anderson Maciel, and Luca Brayda. 2016. Spatial discrimination of vibrotactile stimuli around the head. In *2016 IEEE Haptics Symposium (HAPTICS)*, Vol. 2016-April. IEEE, 1–6. <https://doi.org/10.1109/HAPTICS.2016.7463147>
- [3] Jan B. F. Van Erp, Hendrik A. H. C. Van Veen, Chris Jansen, and Trevor Dobbins. 2005. Waypoint Navigation with a Vibrotactile Waist Belt. *ACM Transactions on Applied Perception* 2, 2 (apr 2005), 106–117. <https://doi.org/10.1145/1060581.1060585>
- [4] Garmin. 2021. Varia RTL515, Radar tail light. Retrieved January 11, 2021 from <https://buy.garmin.com/en-GB/GB/p/698001>
- [5] Wilko Heuten, Niels Henze, Susanne Boll, and Martin Pielot. 2008. Tactile wayfinder: A Non-Visual Support System for Wayfinding. In *Proceedings of the 5th Nordic conference on Human-computer interaction building bridges - NordiCHI '08*. ACM Press, New York, New York, USA, 172. <https://doi.org/10.1145/1463160.1463179>
- [6] Eric M. Jones, Ted Selker, and Hyemin Chung. 2007. What you said about where you shook your head: A hands-free implementation of a location-based notification system. In *CHI '07 extended abstracts on Human factors in computing systems - CHI '07*. ACM Press, New York, New York, USA, 2477. <https://doi.org/10.1145/1240866.1241027>
- [7] Francisco Kiss, Robin Boldt, Bastian Pflöging, and Stefan Schneegass. 2018. Navigation Systems for Motorcyclists. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems*, Vol. 2018-April. ACM, New York, NY, USA, 1–7. <https://doi.org/10.1145/3173574.3174191>
- [8] Jinlong Machinery & Electronics Co. LTD. 2021. KOTL-S08067, Coin Type Vibration Motor product specification. Retrieved February 24, 2021 from <https://www.digkey.co.uk/htmldatasheets/production/1917789/0/0/1/c0834b011f.html>
- [9] Andrii Matviienko, Swamy Ananthanarayan, Shadan Sadeghian Borojeni, Yannick Feld, Wilko Heuten, and Susanne Boll. 2018. Augmenting bicycles and helmets with multimodal warnings for children. *MobileHCI 2018 - Beyond Mobile: The Next 20 Years - 20th International Conference on Human-Computer Interaction with Mobile Devices and Services, Conference Proceedings* (2018), 1–13. <https://doi.org/10.1145/3229434.3229479>
- [10] Andrii Matviienko, Swamy Ananthanarayan, Abdallah El Ali, Wilko Heuten, and Susanne Boll. 2019. NaviBike: Comparing Unimodal Navigation Cues for Child Cyclists. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*. ACM, New York, NY, USA, 1–12. <https://doi.org/10.1145/3290605.3300850>
- [11] Martin Pielot, Benjamin Poppinga, Wilko Heuten, and Susanne Boll. 2012. Tacticycle: Supporting Exploratory Bicycle Trips. In *Proceedings of the 14th international conference on Human-computer interaction with mobile devices and services - MobileHCI '12*. ACM Press, New York, New York, USA, 369. <https://doi.org/10.1145/2371574.2371631>
- [12] Benjamin Poppinga, Martin Pielot, and Susanne Boll. 2009. Tacticycle — A Tactile Display for Supporting Tourists on a Bicycle Trip. In *Proceedings of the 11th International Conference on Human-Computer Interaction with Mobile Devices and Services - MobileHCI '09*. ACM Press, New York, New York, USA, 1. <https://doi.org/10.1145/1613858.1613911>
- [13] Manoj Prasad, Paul Tael, Daniel Goldberg, and Tracy A. Hammond. 2014. HaptiMoto: Turn-by-Turn Haptic Route Guidance Interface for Motorcyclists. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*. ACM, New York, NY, USA, 3597–3606. <https://doi.org/10.1145/2556288.2557404>
- [14] Haska Steltenpohl and Anders Bouwer. 2013. Vibrobelt. In *Proceedings of the 2013 international conference on Intelligent user interfaces - IUI '13*. ACM Press, New York, New York, USA, 417. <https://doi.org/10.1145/2449396.2449450>
- [15] Wouter Walmlink, Alan Chatham, and Florian Mueller. 2014. Interaction opportunities around helmet design. In *CHI '14 Extended Abstracts on Human Factors in Computing Systems*. ACM, New York, NY, USA, 367–370. <https://doi.org/10.1145/2559206.2574803>
- [16] Wouter Walmlink, Danielle Wilde, and Florian Floyd Mueller. 2013. Displaying heart rate data on a bicycle helmet to support social exertion experiences. In *Proceedings of the 8th International Conference on Tangible, Embedded and Embodied Interaction - TEI '14*. ACM Press, New York, New York, USA, 97–104. <https://doi.org/10.1145/2540930.2540970>
- [17] M. Wierda and K. A. Brookhuis. 1991. Analysis of cycling skill: A cognitive approach. *Applied Cognitive Psychology* 5, 2 (mar 1991), 113–122. <https://doi.org/10.1002/acp.2350050205>
- [18] Chao-Yang Yang, Yu-Ting Wu, and Cheng-Tse Wu. 2014. Impact of Multi-sensory On-Bicycle Rider Assistance Devices on Rider Concentration and Safety. In *Design, User Experience, and Usability. User Experience Design for Everyday Life Applications and Services*, Aaron Marcus (Ed.). Springer International Publishing, Cham, 378–388.